

Effect of Segmentation on the Performance of Melanoma Classification Models with Teachable Machine

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Abstract

Early distinction between benign, inflammatory, and malignant skin lesions is crucial for increasing the probability of therapeutic success and patient survival, especially in more severe skin lesions like melanoma. This project will analyze the influence of dermoscopic image segmentation on the performance of skin lesion classification models using the Teachable Machine platform as a machine learning tool. [1].

1 Introduction

The skin is the largest organ in the human body, and over the last few years, there has been a significant increase in the incidence of malignant melanoma [2]. Melanoma, one of the most common types of cancer, is responsible for over 80% of skin cancer mortality in Europe, accounting for 1 to 2% of all annual deaths. It arises when melanocytes (pigment cells) become malignant [3]. Data from the latest National Oncology Registry reveals that skin cancers in Portugal have been gradually increasing (more than 3600 new cases) [4]. The early detection of this type of lesion is, therefore, determining, meaning that when identified in an initial stage, melanoma can show survival rates above 90%; however, this value decreases considerably when diagnosed late [5].

The ABCDE rule is frequently recommended for the melanoma treatment process. Described by Robert Friedman, this rule states that asymmetry (A), border irregularity (B), color variation (C), diameter usually greater than 6mm (D), and evolution (E) — regarding size, shape, color tone, surface characteristics, or symptoms — are characteristics that contribute to the early diagnosis of the disease [3]. Studies indicate that this rule is useful for alerting that something may be wrong with a skin lesion. However, this is insufficient, it alone may not be enough for the diagnosis of pre-invasive melanoma. In order to overcome these limitations, the development of Artificial Intelligence has been crucial. Teachable Machine is a web-based tool with a graphical interface that allows training basic classification models. It contributes to the analysis of skin lesion images by allowing the training of a classification model using examples of melanomas and non-melanomas. Thus, it can function as an initial screening tool, assisting in the early detection of lesions [6].

Additionally, studies demonstrate the usefulness of segmentation in improving the performance of classification based on Deep Learning for certain diagnosed categories [7]. Skin image segmentation (responsible for isolating the lesion from the rest of the skin) significantly improves the quality of the data used in classification models. By integrating this step with Teachable Machine, it is possible to provide the model with more consistent regions focused on the lesion, increasing the learning accuracy, in addition to enhancing the performance of Teachable Machine in distinguishing between melanomas and non-melanomas. And finally, the impact of this dermoscopic image segmentation on the performance of automatic classification models for skin lesions will be analyzed.

2 Methods

As previously mentioned, the main objective of the study is associated with the analysis of the impact of image pre-processing, namely segmentation, on the performance of automatic skin lesion classifiers using dermoscopic images.

Thus, a database consisting of a set of skin lesion images divided into two classes was used: Melanoma and Non-Melanoma. However, due to having different quantities of each class and in order to balance the dataset, it was decided to increase the number of images through data augmentation, creating altered versions of original images, so that each class now contains exactly 80 images. With this, a script was created to implement a set of transformations to the original images, in a random manner. These transformations included horizontal and vertical reflections, small rotations between -10° and 10° , and small translations in X and Y (horizontal and vertical).

Furthermore, the contrast (between 80 and 120%) and brightness (between -0.1 and +0.1) of some images were modified. These changes were applied with the intention of creating varied versions of the original images and thus increasing the variety of the database. In addition, during pre-processing, an output size of 256x256 pixels was defined to ensure that all images have the same dimensions. Finally, a dataset was obtained where both classes contained exactly 80 images, all of the same size and with small visual variations derived from the

originals, promoting a balanced and diversified dataset for training the classification models.

```
imageAugmenter = imageDataAugmenter( ...  
    'RandXReflection', true, ...    % Reflexão horizontal  
    'RandYReflection', true, ...    % Reflexão vertical  
    'RandRotation', [-10, 10], ...  % Rotação aleatória  
    'RandXTranslation', [-3 3], ... % Translação horizontal  
    'RandYTranslation', [-3 3]);    % Translação vertical
```

Figure 1: Matlab code for editing the images

As mentioned earlier, the Teachable Machine platform was used as a tool for training the models, using images as input data in this case. We began by selecting and separating the set into two classes (previously edited images). Where the first class corresponded to images with melanoma and the second class to images without melanoma.

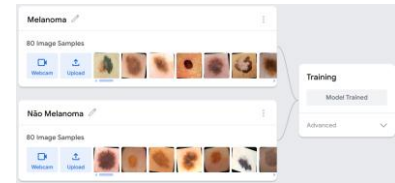


Figure 2: Organization of images into two classes (melanoma and nonmelanoma) on the Teachable Machine platform

In order to meet the objective of the study in question, the images were segmented, isolating the skin lesion. The goal of this procedure is to reduce visual noise and allow for a direct comparison between models trained with original images and segmented images. The segmentation process was carried out using Matlab and follows a sequence of steps that allowed the isolation of the lesion in each image. Generally, a combination of automatic binarization, polarity correction, and hole filling was performed. Below, we can analyze the difference between a non-segmented image and a segmented image, respectively, and confirm the isolation of the lesion.

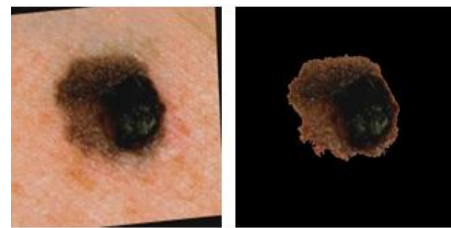


Figure 3: (a) non-segmented image; (b) Segmented image

3 Results and Discussion

After training the respective images, without segmentation, the following results were obtained:

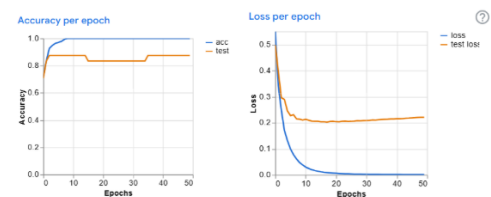


Figure 4: (a)Accuracy per epoch; (b)Loss per epoch

The graphs demonstrate a case of overfitting, for the following parameters (defined by default by the platform): 50 epochs, a batch size of 16, and a learning rate of 0.001. This occurs when the model learns to classify the training samples so closely that it fails to make correct classifications on the test samples; in other words, it limits its ability to generalize to external data that is not part of its respective set. In graph a), one sees a significant and rapid increase (shortly before the 10th epoch) in the blue curve, demonstrating that the model is capable

of classifying all training data perfectly. For the test data, there is also an initial increase that stabilizes around 0.87 (87%), shortly before the 20th epoch. This difference (13%) between test and training Accuracy indicates the model's overfitting.

In graph b), the training loss (blue curve) drops significantly and approaches 0, which may indicate that the model is memorizing the training set. On the other hand, the test loss (orange curve) follows the training drop, but stabilizes around 0.2 and starts to slowly rise again around the 20th epoch. In summary, this point where the test loss stops decaying, stabilizes, and begins to rise slowly indicates that the model has stopped learning generalized patterns and starts learning the noise, leading to overfitting.

Through the confusion matrix, it was concluded that this model has: 11 true positives, 10 true negatives, 2 false positives, and 1 false negative.

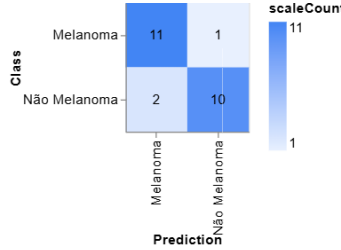


Figure 5: Confusion matrix (model without segmentation)

In addition, it is also possible to calculate the following metrics:

$$Accuracy = \frac{11+10}{11+10+2+1} = \frac{21}{24} \approx 0.875 \approx 87.5\%$$

$$Recall(M) = \frac{11}{11+1} \approx 0.917 \approx 91.7\%$$

$$Recall(NM) = \frac{10}{10+2} \approx 0.833 \approx 83.3\%$$

$$F1_score(M) = 2 \times \frac{0.846 \times 0.917}{0.846 + 0.917} \approx 0.879 \approx 88\%$$

$$F1_score(NM) = 2 \times \frac{0.9091 \times 0.8333}{0.9091 + 0.8333} \approx 0.87 \approx 87\%$$

Accuracy measures the proportion of correct classifications relative to the total number of samples. Recall, in turn, measures the proportion of true cases of a class that were correctly identified (it was calculated for both classes: M-melanoma, NM-non-melanoma). And finally, the F1-score is the average between the two previously calculated metrics (also calculated for both classes). Overall, the obtained results indicate, according to the performance metrics, how efficient the model in question is. However, it is important to take the context of the situation into account. In the scope of cancer diagnosis, the recall of the Melanoma class is the most important, since a false negative can be dangerous and a very serious error. A Recall of 91.7% was obtained for Melanoma, which can be considered excellent. This demonstrates that the model has good performance.

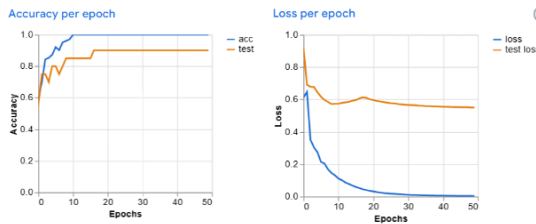


Figure 6: (a) Accuracy per epoch; (b) Loss per epochs

Subsequently, the analysis of the results obtained from the segmentation was carried out using the same parameters: 50 epochs, a batch size of 16, and a learning rate of 0.001. The graphs demonstrate an improvement in the model with the segmented images when compared to the previously analyzed model. In the first graph (a), the training curve (blue) gradually increases, reaching 1.0 (100%) at the 10th epoch, indicating, once again, that the model was able to learn and classify the training data perfectly. In turn, the orange curve shows the test accuracy, which rapidly increases and reaches its maximum (90%) at the 16th epoch. In the second graph (b), the training loss curve (blue) drops abruptly, reaching close to 0 around the 30th epoch. The orange curve (test loss) follows the blue one but stabilizes at approximately 0.55. Contrary to the previous test (non-segmented images), this one does not rise again, remaining stable at 0.55 until the end of the training. This factor may indicate the absence of severe overfitting as in the previous model. Thus, validating the effectiveness of image segmentation before model training.

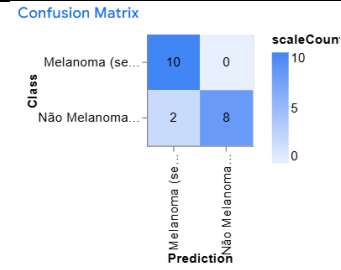


Figure 7: Confusion matrix (model with segmentation)

Similarly, through the confusion matrix, it was concluded that this model has: 10 true positives, 2 false positives, 8 true negatives, and 0 false negatives. It was also possible to calculate the performance metrics for the segmented model:

$$Accuracy = \frac{10+8}{20} = \frac{18}{20} \approx 0.90 \approx 90\%$$

$$Recall(M) = \frac{10}{10+0} \approx 1.00 \approx 100\%$$

$$Recall(NM) = \frac{8}{8+2} \approx 0.80 \approx 80\%$$

$$F1_score(M) = 2 \times \frac{0.8333 \times 1.00}{0.8333 + 1.00} \approx 0.9091 \approx 90.91\%$$

$$F1_score(NM) = 2 \times \frac{1.00 \times 0.80}{1.00 + 0.80} \approx 0.8889 \approx 88.89\%$$

Compared to the previous model (without segmentation), this one showed an improvement in Recall for the Melanoma class, reaching 100% and demonstrating that it did not commit any false negatives. The false positives in this model remain at 2; however, since there was an increase in recall, the false positive is no longer considered as critical in this case.

4 Conclusion

Through the study conducted, and the comparison between the models with segmented and non-segmented images, it was concluded that the pre-processing of images with segmentation was effective to some extent, resulting in a model without overfitting. Furthermore, the *recall* of the segmented model showed a significant improvement relative to the *recall* of the model without segmentation. Additionally, by analyzing the graph, it was observed that the loss (*Loss per epochs*) remained stable, and the accuracy (*Accuracy per epochs*) maintained similar values, for both training and testing. Finally, it is possible to conclude that segmentation positively influenced the performance of automatic classifiers of skin lesions using dermoscopic images.

References

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